Checkpoint B - question paper

Attempt guide

Checkpoint B is a formative cumulative assessment of selected required material from Weeks 5–7. It contains 40 marks and has a 75-minute time limit. All questions and parts are compulsory.

Student sequence

1. Open the formula sheet and candidate paper only.
2. Complete the paper in 75 minutes without notes or solutions.
3. Use the full solutions to review the attempt and identify revision priorities.
4. Record the first cause of each lost mark by learning-outcome ID.
5. Revisit the named Week 5, 6 or 7 material and reattempt missed parts from a blank page after two or three days.

Attempt sequence

1. Open the question paper and formula sheet.
2. Complete the paper in 60 minutes without notes or solutions.
3. Use the full solutions to review the attempt and identify revision priorities.
4. Record the first cause of each lost mark and schedule a blank-page reattempt after two or three days.

Conditions

A basic non-programmable scientific calculator is permitted, although no question requires one. Software, programmable calculators, computer algebra and statistical packages are not permitted. The checkpoint contains no optional dynamics, regression or other excluded content.

Checkpoint B is diagnostic and not pass/fail. Use the evidence from the attempt to guide revision and reattempts. For this controlled first-cohort pilot, scores are not used for admission, progression, ranking, funding or grading.

Candidate paper

Time allowed: 75 minutes
Total marks: 40
Questions: All questions and all parts are compulsory.

Instructions

- Complete the paper without notes or solutions.
- A basic non-programmable scientific calculator is permitted, but no question requires one.
- Software, programmable calculators, computer algebra and statistical packages are not permitted.
- Show the reasoning and intermediate work needed to support each answer. An unsupported answer may not receive full marks.
- Give exact answers unless the question asks for an approximation.
- All vectors are column vectors. Use the course notation throughout.
- For inequality-constrained maximisation, write every constraint as $g_j(\mathbf{x}) \geq 0$ and use the course Kuhn–Tucker convention.

Question 1 — quadratic forms and convexity (4 marks; 8 minutes)

Part A — `CPB-Q01-PA` (4 marks; 8 minutes) Let

$$\mathbf{A} = \begin{bmatrix} -2 & 3 \\ -1 & -1 \end{bmatrix}, \qquad q(\mathbf{x}) = \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}, \qquad \mathbf{x} \in \mathbb{R}^2.$$

Find the symmetric matrix that represents the same quadratic form and classify q by definiteness using a valid test. Hence determine whether

$$C := \{\mathbf{x} \in \mathbb{R}^2 : q(\mathbf{x}) \geq -1\}$$

is convex. Justify the conclusion without relying on a sketch.

Question 2 — unconstrained optimisation (5 marks; 8 minutes)

Part A — `CPB-Q02-PA` (5 marks; 8 minutes) For

$$f(x, y) = 6x + 8y - 2x^2 - y^2, \qquad (x, y) \in \mathbb{R}^2,$$

find every stationary point and classify it. State the condition that turns the local classification into a unique global conclusion on the stated domain, then calculate the global optimum value.

Question 3 — equality-constrained optimisation and the envelope theorem (8 marks; 15 minutes)

For a resource parameter $a > 0$, consider

$$\max_{x>0,\, y>0} \sqrt{xy} \quad \text{subject to} \quad 2x + y = a.$$

Encode the constraint as $a - 2x - y = 0$.

Part A — `CPB-Q03-PA` (5 marks; 10 minutes) Form the Lagrangian and solve the first-order and feasibility conditions for $x^*(a)$, $y^*(a)$ and the multiplier $\mu^*(a)$. State a sufficient condition for the solution to be the unique global maximiser. Report the optimisers and multiplier at $a = 12$.

Part B — `CPB-Q03-PB` (3 marks; 5 minutes) Derive the value function $V(a)$. Use the envelope theorem to calculate $V'(a)$, verify the result by direct differentiation of $V(a)$, and interpret its sign under the stated constraint encoding.

Question 4 — a Kuhn–Tucker corner (6 marks; 13 minutes)

Part A — `CPB-Q04-PA` (6 marks; 13 minutes) Consider

$$\max_{x,y} \{4x + y - x^2 - y^2\}$$

subject to

$$x \geq 0, \qquad y \geq 0, \qquad 1 - x - y \geq 0.$$

Form the Lagrangian and write the complete Kuhn–Tucker conditions under the course convention. Solve them, explicitly identifying the active and slack constraints and reporting every multiplier. Use curvature and the feasible set to justify whether the maximiser is unique and global.

Question 5 — a joint density and conditional expectation (5 marks; 10 minutes)

Part A — `CPB-Q05-PA` (5 marks; 10 minutes) The joint density of X and Y is

$$f_{X,Y}(x, y) = x + y, \qquad 0 < x < 1, \quad 0 < y < 1,$$

and is zero otherwise.

Derive $f_Y(y)$ and $f_{X|Y}(x \mid y)$, including their supports. Then calculate $\mathbb{E}[X \mid Y = y]$ and determine whether X and Y are independent. Justify the independence conclusion.

Question 6 — exact and approximate sampling distributions (5 marks; 9 minutes)

Part A — `CPB-Q06-PA` (5 marks; 9 minutes) Let

$$X_1, \dots, X_{32} \overset{\text{iid}}{\sim} \text{Poisson}(2),$$

where 2 is the mean and rate, and define

$$T := \sum_{i=1}^{32} X_i, \qquad \bar{X} := \frac{1}{32} \sum_{i=1}^{32} X_i.$$

State the exact distribution of T, and calculate $\mathbb{E}[\bar{X}]$ and $\text{Var}(\bar{X})$. Then use the central limit theorem, without a continuity correction, to approximate $\mathbb{P}(\bar{X} > 2.5)$. You may use $\Phi(2) = 0.9772$, where Φ is the standard normal CDF. Explain why the distributional statement for T is exact but the final probability calculation is approximate.

Question 7 — method of moments and maximum likelihood (7 marks; 12 minutes)

Part A — `CPB-Q07-PA` (7 marks; 12 minutes) Let

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Geometric}(p), \qquad 0 < p \leq 1,$$

where X_i is the number of trials up to and including the first success, so its support is $\{1, 2, ...\}$, its PMF is

$$p_X(x) = p(1-p)^{x-1}, \qquad x \in \{1, 2, ...\},$$

and $\mathbb{E}_p[X_i] = 1/p$.

Derive the one-moment method-of-moments estimator of p. Derive the maximum-likelihood estimator, treating both samples with $\sum_{i=1}^{n} x_i > n$ and the boundary sample $x_1 = \cdots = x_n = 1$. Explain the relationship between the two estimators. Calculate the estimate for the observed sample $(1, 2, 1, 4)$, and state why the estimator is consistent for p.

Formula sheet

General conventions

- Give exact answers unless a question asks for an approximation.
- All vectors and gradients are column vectors.
- For a maximisation problem, write inequalities as $g_j(\mathbf{x}) \geq 0$ and use the course Kuhn–Tucker sign convention stated in the course materials.
- For a random sample, $\bar{X} = n^{-1} \sum_{i=1}^{n} X_i$.
- Φ denotes the standard normal cumulative distribution function.

Formula provision

The geometric PMF, support and mean needed in Question 7 and the numerical normal CDF value needed in Question 6 are supplied in the candidate paper. No additional formulae or numerical tables are supplied because the definiteness, optimisation, conditioning, sampling and estimation identities used in the questions are part of the assessed material.